

BIO503 BIOPHYSICS M.C.Q.S MEGA FILE

ALL DATA FROM QUIZES + GRAND QUIZES

GROUP: BIOTECH BRAINY BUNCH

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Grand quiz 2020

1. The LDF is named after the German-American physicist? **(Fritz London)**
2. Device that measure pressure using mechanical deformation. **(Bourdon Tube)**
3. The cell membranes are involved in a variety of cellular processes such as ____ **(cell adhesion, ion conductivity, cell signaling, All)**
4. What heat flows through a heated solid. **(Conduction)**
5. Reaction with a positive ΔG ($\Delta G > 0$). On the other hand require an input of energy and are called ____ **(Endergonic reaction)**
6. The most important energy transfer event, for life on our planet is the biological process called ____ **(Photosynthesis)**
7. The first National Biophysics Congress was held in ____ **(1963)**
8. Muscle contraction is done by ____ **(Myosine)**
9. Atmospheric pressure is close to ____ **(100000 Pascal)**
10. ____ is the study of the electrical nature of tissues and cells. **(Electrophysiology)**
11. ____ does not require a cell to use energy. **(Passive transport)**
12. In a two phase system, and when the mass of each phase reaches an equilibrium level and stays there. **(Phase equilibrium)**
13. It is process during which the specific volume remains constant. **(Isochoric)**
14. The third law of thermodynamics was developed by the ____ **(Walther Nernst)**
15. The enthalpy change required to complete change the state of one mole of substances between solid and gaseous states ____ **(Enthalpy of sublimation)**
16. An ____ is a system that freely exchanges energy and matter with its surroundings. **(Open system)**

17. _____ scanners use strong magnetic field, electric field gradients, and radio waves to generate images of the organs in the body____ **(MRI)**
18. RNA is distributed throughout the cell, most commonly I small numerous organelles Called____ **(Ribosomes)**
19. _____ is a green pigment required for photosynthesis. **(Chlorophyll)**
20. If there is no change in pressure at any point of the system with time. **(Mechanical equilibrium)**
21. A calorimeter is a device for measuring_____ **(Transfer of heat)**
22. _____ pressure is defined as the pressure exerted by a vapour in thermodynamics equilibrium with its condensed phases (solid or liquid) at a given temperature in a closed system. **(Vapour)**
23. It is a process during which the temperature remains constant. **(Isothermal)**
24. The field of physics that deals with the relationship between heat and other properties (such as pressure, density, temperature, etc.) in a substance. **(Thermodynamics)**
25. The average bond energy of O-H bonds in water is ____ kJ/mol. **(463)**
26. Electrophysiology study is also known as? **(EP Test)**
27. Physics provides the fundamental theories for understanding_____. **(Bio-molecules)**
28. When two substances can affect each other's temperature. **(Thermal Contact)**
29. The scientific study of the motion of fluids, especially non-compressible liquids, under the influence of internal and external forces is known as? **(Hydrodynamics)**
30. Non polar covalent compounds like methane CH_4 , and nitrogen gas N_2 only have ____ between molecules. **(London dispersion forces)**
31. The_____ groups of these amino acids have either aliphatic or aromatic groups. This makes them hydrophobic ("water fearing"). **(R group)**
32. The energy required to separate one mole of an ionic compound into separated gaseous ions to an infinite distant apart. **(lattice enthalpy)**
33. Device that measure atmospheric pressure. It is a manometer with a near vacuum on one end____ **(Barometer)**
34. An ion-dipole force is an attractive force that results from the electrostatic attraction between an ion and a _____ molecule that has a dipole. **(neutral)**
- 35.
36. Size of proteins I between _____ **(5 to 10 nm)**
37. Measure of degree of disorder or randomness in a molecular system is called_____ **(Entropy)**
38. What protects body from foreign pathogens **(Antibodies)**
39. The energy required to break a bond_____ **(bond energy)**
40. _____ is the totality of an organism's chemical reactions. **(Metabolism)**
41. The _____ of a system is equal to the system's internal energy plus the product of its pressure and volume. **(Enthalpy)**

42. Any change that a system undergoes from one equilibrium state to another is called a _____ **(Process)**
43. When a low conducting material is used to prevent heat transfer. **(Insulation)**
44. A _____ is a system that exchanges only energy with its surrounding, no matter. **(Closed system)**
45. _____ consumes energy to build complex molecules from simpler ones. **(Anabolic pathway)**
46. When heat flows through a heated solid. **(conduction)**
47. Adenine and guanine are _____, meaning that require their structure contain two fused carbon-nitrogen rings **(purines)**
48. Ionic bonding is a type of chemical bond involves the electrostatic attraction between _____ charged ions. **(oppositely)**
49. Metallic bonding are the force of attraction between _____ electrons and metal atom. **(valence)**
50. Freezing point of water in Fahrenheit is _____ **(32°F)**
51. _____ is the study of the physical principles governing biological membranes, including lipid-raft formation and protein- lipid coupling. **(Membrane Biophysics)**
52. Boiling point of water in Fahrenheit _____ **(212°F)**
53. When a protein losses its high order structure, but not its primary sequence, it is said to be _____. **(denature)**
54. The building blocks of proteins are called _____. **(Amino acids)**
55. The energy that a system possesses as a result of its elevation in a gravitational field. **(potential energy)**
56. The term biophysics was originally introduced by _____ in 1892. **(Karl Pearson)**
57. In eukaryotes, such as plants and animals, DNA is found in the **(Nucleus)**
58. What makes molecules to attach to one another? **(Intermolecular forces)**
59. _____ releases energy by breaking down complex molecules into simpler compounds. **(Catabolic pathway)**
60. Twenty percent of the human body is made up of _____. **(proteins)**
61. Hemoglobin is a protein found in the _____, and its function is to carry oxygenated blood to various part of the body. **(RBCs)**
62. The ability of a substance to be beaten into thin sheets. **(Malleability)**
63. Biophysics discovers how atoms are arranged to work in _____. **(DNA & proteins)**
64. DNA was discovered in _____. **(1953)**
65. Synthesis of RNA is usually catalyzed by an enzyme called. **(RNA Polymerase)**
66. Foundation of Italian society for Fundamental and Applied Biophysics was in _____. **(1973)**
67. The society publishes the biophysical journal _____ monthly. **(twice)**
68. Size of small fatty acids and sugars is _____. **(1nm)**
69. In eukaryotes, such as plants and animals, DNA is found in the **(Nucleus)**

70. Macromolecules formed from joining together of small organic molecules. **(Biological Polymers)**
71. When one mole of gaseous ions are completely dissolved in water forming one mole of aqueous ions **(Enthalpy of hydration)**
72. The energy that a system possesses as a result of its motion relative to some reference of frame. **(Kinetic energy)**
73. The force of attraction between a non-polar molecule and an ion that may lie in its vicinity is called? **(ion-induced dipole forces)**
74. The American Biophysics Society was founded in? **(1957)**
75. _____ is the process in which cells absorb molecules by engulfing them. **(Endocytosis)**
76. When heat is transferred through electromagnetic waves, such as from sun. **(Radiation)**
77. Electrophysiology study is used to study abnormal? **(Heart rhythms)**
78. Intermolecular forces are typically _____ than intramolecular forces, and account for the bulk properties of matter (e.g., boiling point, melting point, etc.) **(Weaker)**
79. Without _____, earth would have no water cycle, no ocean, no floating ice, no life as we know it. **(hydrogen bonding)**
80. The ability of substances to be drawn into thin wires. **(Ductility)**
81. The fluid mosaic model describes the structure of? **(Plasma membrane)**
82. In prokaryotes, such as bacteria, the DNA is not enclosed in a membranous envelope, although it's located in a specialized cell region called the _____ **(Nucleoid)**
83. There were two major categories of proteins with quaternary structure _____ **(Fibrous and Globular)**
84. The ability of a substance to allow the flow of energy of electricity. Metals like copper, bronze, steel, are good conductor of heat and electricity. **(Conductivity)**
85. The unit of measurement of enthalpy in the international system of Units (SI) is the _____ **(Joule)**
86. Metallic bonding is the force of attraction between _____ electrons and the metal atom. **(Valence)**
87. The enthalpy change required to completely change the state of one mole of substance between liquid and gaseous states. **(Enthalpy of Vaporization)**
88. When heated particles heat another substances, such as cooking something in boiling water. **(Convection)**
89. When was series of conference on Biophysics was organized in Genoa _____ **(1959)**
90. Biophysicists discovered the DNA double helix in (1953)

Bio-503 Quiz (01) 2021

1. Experiments in the 1940's showed that genes are made of simple chemical _____ **(DNA)**
2. THE American Biophysics was founded in? **(1957)**

3. The term ____ was originally introduced by Karl Pearson in 1892 **(Biophysics)**
4. The first National Biophysics Congress was held from 3-5 June ____ **(1963)**

(Note: some were repeated present above)

Bio-503 Quiz 2022

1. Parts of your eyes, ears, nose, and skin that senses your environment are made by ____ **(proteins)**

(Note: many are from 2020 quiz)

Bio-503 Quiz

2. ____ quantifies proteins and peptides in solution if primary amine are present in the amino acids. **(Fluoresc-amine)**
3. The speed of dialysis might be increased by using ____ water. **(Hot)**
4. The amino acids are identified by the comparison ____ times of the components in the specimen to those of reference compounds. **(Retention)**
5. ____ is used to transport salt ions from one solution through ion-exchange membranes to another solution under the influence of an applied electric potential difference. **(Electro dialysis)**
6. The ____ is suitable for the determination of the concentration of various antigens, hormones, and drugs in different body fluids such as the human blood. **(Radioimmunoassay)**
7. ____ for a specific protein are used as ligands on an affinity chromatography column. **(Antibodies)**
8. In case of antibody-antigen the inverted dissociation constant is used and is called ____ constant. **(Affinity)**
9. ____ is a natural phenomenon by which all things with mass are brought towards (or gravity toward) one another, including ranging from atoms and photons, to planets and stars. **(Gravity)**
10. ____ is the technique of precipitating a protein antigen out of solution using an antibody that specially bonds to that particular protein. **(protein immune precipitation)**
11. Ninhydrin can react with ____ and give purple colored product. **(alpha amino acid)**
12. ____ are cylindrical polymer and made of α and β tubulents. **(Microtubules)**
13. The inverse of association is called ____ constant. **(Dissociation)**
14. The basic operation principle of ultrafiltration uses a pressure induced separation of solute od solute from a solvent through a ____ **(Semi permeable)**
15. ____ is the random, thermal movement of molecules in the solution that leads to the net movement of molecules from an area of higher concentration to a lower concentration until equilibrium is reached. **(Diffusion)**

16. **Biophysics** is an interdisciplinary science that approaches and methods of physics to study biological system.
17. Biophysics are driven primarily by their curiosity about how biological systems work at **molecular level**.
18. Biophysics is a **molecular science** that seeks to explain biological function in terms of the molecular structures and properties of specific molecules.
19. Who used the X-ray crystallographic, which revealed the physical patterns of DNA? **(Francis Crick)**
20. **James & Watson** biophysicists discovered the DNA double helix in 1953.
21. ____ first developed hypothesis about the physical basis for consciousness and perception. **(Greek and Romans)**
22. MRI stand for ____ **(Magnetic Resonance Imaging)**
23. ____ used in radiology to form pictures of the **anatomy** and the **physiological** processes of the body in both health and diseases. **(Magnetic Resonance Imaging)**
24. Study of protein conformation and orientation. **(Fluorescence)**
25. **Viscosity is the measurement of how well substances flow.**
26. A **covalent**, is also called a **molecular bond**, is a chemical bond that involves the sharing of electron pairs between between atoms.
27. Which of the following are found in liquid state at room temperature? **(Bromine)**
28. The five-carbon sugar in DNA is called deoxyribose, while in RNA, the sugar is **(Ribose)**
29. **Absolute Zero** is denoted as 0 k on the kelvin scale, -273.15°C on the Celsius scale and - 459.67°F on the Fahrenheit scale.
30. **Competitive inhibitors** bind to the active site of an enzyme, competing with the substrate.
31. Which one is not found in the membrane of plant cell? **(Cholesterol)**
32. How many types of friction? **(Four)**
33. **Theodore Svedberg** invented the analytical ultracentrifuge in 1923, and won the Nobel Prize in Chemistry in 1926 for his research on colloids and proteins using ultracentrifuge.
34. In **Rotatory motion**, the particles of the body describe concentric circles about axis of motion.
35. **Static friction** exist between a stationary object and the surface on which it is resting.
36. The substance which reduces friction called? **(lubricants)**
37. ____ are major component of cell membrane? **(Phospholipids)**
38. Lipids can make up anywhere from **20 to 80** percent of the membrane.
39. ICRP stands for ____ **(International committee on Radiation protection)**
40. More than ____ of man-made radiation received by human comes from medical diagnosis. **(90%)**
41. ____ are neutral particles that have no charge. **(Neutrons)**
42. Radiations arising from human activities typically accounts for about ____ of public exposure. **(20%)**

43. X-Ray widely used for medical diagnosis since its discovery in ____ (1895)
44. Coal is an impure fuel, and usually contains ____ of Uranium. (1.3ppm)
45. ____ is produced by the Buthus Tamulus (Eastern Indian Scorpion) and blocks potassium channels. (Iberiotoxin)
46. In Paramyotonia the ____ channels fails to regulate the flow of ions properly. (Sodium)

More Quizzes from Past Papers

1. (Biophysics) is an interdisciplinary science that applies the approaches and methods of physics to study biological systems.
2. Biophysics discovers how atoms are arranged to work in (DNA and proteins).
3. Biophysicists are driven primarily by their curiosity about how biological systems work at the (molecular level).
4. Biophysics is a (molecular science) that seeks to explain biological function in terms of the molecular structures and properties of specific molecules.
5. Who used the X-ray crystallography, which revealed the physical patterns of DNA? (Francis Crick)
6. (James & Watson) biophysicists discovered the DNA double helix in 1953.
7. The (fluid mosaic model) describes the structure of the plasma membrane as a mosaic of components—including phospholipids, cholesterol, proteins, and carbohydrates—that gives the membrane a fluid character.
8. (Membrane biophysics) is the study of the physical principles governing biological membranes, including lipid-raft formation and protein–lipid coupling, as well as their mechanical characteristics.
9. _____ first developed hypotheses about the physical basis for consciousness and perception. (Greeks and Romans)
10. The term biophysics was originally introduced by (Karl Pearson in 1892)
11. The American Biophysics Society was founded in (1957)
12. The first National Biophysics Congress was held from (3-5 June, 1963 in Italy).
13. (Electrophysiology) is the study of the electrical nature of tissues and cells.
14. _____ is study abnormal heart rhythms (arrhythmia). Electrophysiology study (EP Test)
15. The scientific study of the motion of fluids, especially non-compressible liquids, under the influence of internal and external forces is called _____ (Hydrodynamics)
16. _____ used measure the mechanical properties of living material (such as tissue or cells) (Atomic Force Microscopy)
17. MRI stands for _____ (Magnetic Resonance Imaging)

18. _____ used in radiology to form pictures of the anatomy and the physiological processes of the body in both health and disease. (**Magnetic Resonance Imaging**)
19. _____ is the study of protein conformation and orientation. (**Fluorescence**)
20. (**Intramolecular forces**) are the forces that hold atoms together within a molecule.
21. Intermolecular forces are also known as van der Waals forces, named after **Johannes van der Waals** who first postulated them.
22. The average bond-energy for O-H bonds in water is **463 kJ/mol**
23. **Ionic bonding** is a type of chemical bond that involves the electrostatic attraction between oppositely charged **ions**.
24. A **covalent bond**, also called a molecular bond, is a chemical bond that involves the **sharing of electron pairs** between atoms.
25. Metallic bonding is the force of attraction between **valence electrons** and the **metal atoms**.
26. An **ion-induced dipole** attraction is a weak attraction that results when the approach of an ion induces a dipole in an atom or in a **nonpolar molecule** by disturbing the arrangement of electrons in the nonpolar species.
27. The force of attraction between a **non-polar molecule** and an ion that may lie in its vicinity is called ion-induced dipole force.
28. An ion-dipole force is an attractive force that results from the electrostatic attraction between an **ion and a neutral molecule** that has a dipole
29. The force of attraction between a polar molecule and an ion that may lie in its vicinity is called an **ion-dipole force**.
30. Without **hydrogen bonding**, earth would have no water cycle, no ocean, no floating ice, and no life.
31. Hydrogen bonding can create a long chain of water molecules which can overcome the **force of gravity** and travel up to the high altitudes of leaves.
32. The **London dispersion force** is the weakest intermolecular force.
33. _____ is sometimes called an induced dipole-induced dipole attraction. **London dispersion force**
34. The LDF (**London dispersion force**) is named after the German-American physicist Fritz London.
35. Which of the following are found in liquid state at room temperature? **Bromine**
36. Nonpolar covalent compounds—like **methane CH₄**, and nitrogen gas **N₂**- only have London dispersion forces between molecules.
37. **Viscosity** is a measure of how well substances flow.
38. Surface tension is a measure of the **toughness** of the surface of a liquid.
39. **Nucleic acids** are molecules that allow organisms to transfer genetic information from one generation to the next.

40. Nucleic acids are composed of **nucleotide** monomers linked together. Nucleotides contain three parts.
41. The five-carbon sugar in DNA is called **deoxyribose**, while in RNA, the sugar is ribose.
42. Which one of the following is the most important of all biomolecules? **Nucleic acid**
43. RNA nucleotides may also bear adenine, guanine and cytosine bases, but instead of thymine they have another pyrimidine base called **Uracil (U)**.
44. Nuclein was discovered by **Friedrich Miescher in 1869**.
45. In **1938 Astbury and Bell** published the first X-ray diffraction pattern of DNA.
46. In **1953 Watson and Crick** determined the structure of DNA.
47. In eukaryotes, such as plants and animals, DNA is found in the **nucleus**, as well as in certain other types of organelles.
48. In prokaryotes, such as bacteria, the DNA is not enclosed in a membranous envelope, although it's located in a specialized cell region called the **nucleoid**.
49. DNA is a polymer of the four nucleotides **A, C, G, and T**, which are joined through a backbone of alternating phosphate and deoxyribose sugar residues.
50. RNA, is distributed throughout the cell, most commonly in small numerous organelles called **ribosome**.
51. **Ribonucleic acid (RNA)** is a polymeric molecule essential in various biological roles in coding, decoding, regulation, and expression of genes.
52. Synthesis of RNA is usually catalyzed by an enzyme—RNA polymerase using DNA as a template, a process known as **transcription**.
53. **Messenger RNA (mRNA)** is the RNA transcript or RNA copy of the DNA message produced during DNA transcription.
54. **Transfer RNA (tRNA)** has a three dimensional shape and is necessary for the translation of mRNA in protein synthesis.
55. **Ribosomal RNA (rRNA)** is a component of ribosomes and is also involved in protein synthesis.
56. **Twenty percent** of the human body is made up of protein.
57. The covalent bond between the amino acids is called a **peptide bond**.
58. Humans have **tens of thousands** of different proteins, each with their own structure and function.
59. All protein polymers are constructed from the same set of 20 monomers, called **amino acids**.
60. **Enzymes** act as catalysts in biochemical reactions, meaning that they speed the reactions up.
61. Hormones are long-distance chemical signals released by **endocrine cells** (like the cells of your pituitary gland).
62. **Insulin** is an important peptide hormone that helps regulate blood glucose levels.

63. The most common types of secondary structures are the **α helix** and the **β pleated sheet**.
64. When a protein loses its higher-order structure, but not its primary sequence, it is said to be **denatured**.
65. **Thermodynamics** is the field of physics that deals with the relationship between heat and other properties (such as pressure, density, temperature, etc.) in a substance.
66. **Radiation** is when heat is transferred through electromagnetic waves, such as from the sun.
67. Which one is the intensive property of the system? **Temperature, Pressure and Density**
68. A **system** is defined as a quantity of matter or a region in space chosen for study.
69. The SI unit for pressure is **pascals (Newton per square meter)**
70. **Hydrostatic pressure** is the stress, or pressure, exerted equally in all directions at points within a confined fluid (liquid or gas).
71. A device that measures pressure using a column of liquid is called a **Manometer**.
72. **Bourdon** Tube is a device that measures pressure using mechanical deformation.
73. **Barometer** is a device that measures atmospheric pressure.
74. The modern unit of heat (and work) is the Joule: **1 cal = 4.184 J**.
75. The energy that a system possesses as a result of its motion relative to some reference frame. **Kinetic Energy**
76. The sum of all the microscopic forms of energy is called the **internal energy**.
77. The amount of energy required to raise the temperature of an object by 1 K (1°C) is its **heat capacity**.
78. The mathematical relation for the 1st Law of thermodynamic____ **$\Delta U = Q - W$**
79. The enthalpy of a system is equal to the system's internal energy plus the product of its **pressure and volume**.
80. **Lattice enthalpy** is a measure of the strength of the forces between the ions in an ionic solid.
81. Measure of degree of Disorder or randomness in a molecular system is called **ENTROPY. Units: J/K/mol (Joules per Kelvin per mole)**.
82. The third law of thermodynamics was developed by the German chemist **Walther Nernst** during the years 1906–12.
83. **Absolute zero** is denoted as 0 K on the Kelvin scale, **-273.15 °C** on the Celsius scale, and -459.67 °F on the Fahrenheit scale.
84. If two bodies are each in thermal equilibrium with some third body, then they are also in equilibrium with each other is called **Zeroth Law of Thermodynamics**.
85. The free energy of a system is the sum of its **enthalpy (H)** plus the product of the temperature (Kelvin) and the entropy (S) of the system.
86. Reactions that have a negative ΔG release free energy and are called **exergonic reactions**.

87. A sequence of chemical reactions, where the product of one reaction serves as a substrate for the next, is called a **metabolic pathway or biochemical pathway**.
88. **Catabolic pathways** release energy by breaking down complex molecules into simpler compounds.
89. The most important transfer event, for life on our planet is the biological process called **photosynthesis**, which occurs in most plants, algae, and some bacteria.
90. Free energy is important to cells (organisms) and cells store and release free energy using the **ATP** molecule.
91. All reactions, both endergonic and exergonic, require an input of energy to get started. This energy is called **activation energy**.
92. **Metabolism** is an emergent property of life that arises from interactions between molecules within the cell.
93. Hydrolysis of sucrose by the enzyme **sucrase** is an example of an enzyme-catalyzed reaction.
94. An organic cofactor is called a **coenzyme**. Coenzymes include vitamins.
95. **Competitive inhibitors** bind to the active site of an enzyme, competing with the substrate.
96. A **Cell membrane** is a thin semi-permeable membrane that surrounds the cytoplasm of a cell, enclosing its contents.
97. The cell membrane is a selectively permeable membrane that is composed of **hydrophilic heads and hydrophobic tails**.
98. Some molecules or particles are just too large to pass through the plasma membrane or to move through a **transport protein**.
99. **Endocytosis** is the process in which cells absorb molecules by engulfing them.
100. **Active transport** which requires a cell to use energy to move molecules through its cell membrane.
101. Lipids can make up anywhere from **20 to 80** percent of the membrane.
102. While lipids help to give membranes their **flexibility**, proteins monitor and maintain the cell's chemical climate.
103. Which one of the following is the basic unit of Life? **Cell**
104. Which model describes the structure of cell membrane? **Fluid mosaic model**
105. _____ are a major component of cell membranes. **Phospholipids**
106. Which one is not found in the membranes of plant cells? **Cholesterol**
107. **Glycolipids** are located on cell membrane surfaces and have a carbohydrate sugar chain attached to them.
108. **Peripheral membrane** proteins are exterior to and connected to the membrane by interactions with other proteins.
109. **Integral membrane** proteins are inserted into the membrane and most pass through the membrane.

110. However, under certain circumstances, oxygen atoms bond into triplets (O₃), forming a molecule known as **ozone**.
111. **Molecular chemistry** deals with the laws governing the interaction between molecules that result in the formation and breakage of chemical bonds.
112. _____ a substance tends to move from an area of high concentration to an area of low concentration until its concentration becomes equal throughout a space.
Diffusion
113. In mammals, birds, reptiles and amphibian, **oxygen and carbon dioxide** are exchanged by diffusion in the lungs.
114. **Facilitated diffusion** is the passive movement of molecules across the cell membrane via the aid of a membrane protein.
115. Carrier proteins may move molecules against concentration gradients in the presence of **ATP**
116. Bundles of axons provide a barrier to perpendicular diffusion and a path for parallel diffusion along the orientation of the fibers. This is termed **anisotropic diffusion**.
117. In higher green plants, some substances enter the living cells through the aerial organs—the diffusion of **CO₂ and O₂** from the atmosphere into the plants is principally through stomata.
118. Permeable membranes are those which allow diffusion of either **solvent and solute molecules** or ions through them.
119. **Semi permeable or differentially permeable** membranes are those which allow the diffusion only of the solvent particles through them while restricting the entrance of the solute particles.
120. Nerve cells called **neurons** communicate with other cells by sending electrical signals along their cell membrane.
121. _____ is a type of diffusion that involves the random movement of water molecules. **Osmosis**
122. The process of regulating the concentration of water and mineral salts in the blood is called **Osmoregulation**.
123. _____ is a mathematical object, known as random process that describes a path that consists of a succession of random steps on some mathematical space such as the integers. **Random walk**
124. The term random walk was first introduced by **Karl Pearson in 1905**.
125. In **1827, Robert Brown** was looking through a microscope at pollen grains in water.
126. How many types of friction? **Four**
127. **Static friction** exists between a stationary object and the surface on which it is resting.
128. The substance which reduces friction are called **lubricants**.
129. First compiled by Sir Isaac Newton in his work Philosophiæ Naturalis Principia Mathematica, first published on **July 5, 1687**.

130. The Milky Way Galaxy is moving through space and many astronomers believe the velocity of this motion to be approximately **600 kilometers per second**.
131. An Earth complete orbit around the sun takes one year, or about 365 days; it averages a speed of about **30 kilometers per second (67,000 mph)**.
132. In **Rotatory motion**, the particles of the body describe concentric circles about the axis of motion.
133. **Electrophoresis** to separate molecules of different molecular mass/size based on the sedimentation principle; electric field acts on the charged molecules gel electrophoresis.
134. **Theodore Svedberg** invented the analytical ultracentrifuge in 1923, and won the Nobel Prize in Chemistry in 1926 for his research on colloids and proteins using the ultracentrifuge.

MORE QUIZZES

1. Biophysics is an interdisciplinary science that applies the approaches and methods of _____ to study biological systems. **(Physics)**
2. Biophysical research shares significant overlap with _____
 - a) Biochemistry and physical chemistry
 - b) Nanotechnology and bioengineering
 - c) Computational biology. Biomechanics and system biology
 - d) All of the above.**
3. Biophysics covers all scales of biological organization, from molecular to **(organisms and populations)**
4. Biophysics discovers how atoms are arranged to work in _____ **(DNA and Protein)**
5. _____ are discovering how proteins work, they use these protein structures for discovering how biological machines work, in health and also in disease **(Biophysicists)**
6. _____ study life at every level, from atoms and molecules to cells, organism, and environment **(Biophysicist)**
7. Understanding of by studying fundamental physical principles and mechanisms by which living organism's life to **(Survive, Adapt, Grow)**
8. Fundamental experimental work in areas such as _____. **(molecular structure and dynamics, photosynthesis, cell membrane)**
9. _____ is crossroad where all the interesting sciences interest. **(Biophysics)**
10. Biophysics is a molecular science that seeks to explain biological function in term of the _____ **(molecular sequence, properties of specific molecules)**
11. A technique known as _____ which revealed the physical patterns of DNA. **(X-ray crystallography)**
12. Experiments in the _____ showed the genes are made of a simple chemical DNA **(1940's)**

13. Biophysicists discovered the DNA double helix in **(1953)**
14. During the _____ biophysical inventions decoded all the genes in a human being **(2000's)**
15. All the genes of nearly 200 different species, and some gene from more than _____ other species have been determined. **(100,000)**
16. The term biophysics was originally introduced by Karl Pearson in **(1892)**
17. _____ involves the movement and measurement of voltage and current coursing through the tissues through the ion channel proteins **(Electrophysiology)**
18. There are _____ kinds of forces, or attractions that operate in a molecule **(two)**
19. The American biophysical society was founded in _____ **(1957)**
20. A series of conferences on Biophysics was organized in Genera and in other cities in _____ **(1959)**
21. The first National Biophysics congress was held from 3-5 June, 1963 in **(Italy)**
22. Electrophysiology study is used study abnormal **(Heart rhythms)**
23. The scientific study of the motion of fluids, especially non-compressible liquids, under the influence of internal and external forces is called **(Hydrodynamics)**
24. _____ scanners use strong magnetic fields, electric field gradients, and radio waves to generate images of the organs in the body **(MRI)**
25. Intermolecular forces are typically weaker than intermolecular forces, and account for the bulk properties of matter such that **(boiling point, melting point)**
26. Ionic bonding is a type of chemical bond that involves the electrostatic attraction between _____ charged ions. **(Oppositely)**
27. Metallic bonding is the forces of attraction between _____ electrons and the metal atoms. **(Valence)**
28. Since the hydrogen donor is strongly _____ it pulls the covalently bonded electron pair closer to the nucleus, and away from the hydrogen atom? **(electronegative)**
29. In DNA base A and T are held together by _____ hydrogen bonds **(Two)**
30. In DNA bases G and C are bonded together by these hydrogen bonds **(Three)**
31. Hydrogen bonding is present abundantly in the secondary structure of proteins, and also sparingly in _____ conformation **(Tertiary)**
32. The _____ dispersion forces is the weakest intermolecular force **(London)**
33. _____ force is sometimes called as induced dipole-induced dipole attraction **(London)**
34. The LDF is named after German- American physicist _____ **(Fritz London)**
35. ICI and Br₂ have same masses _____ and therefore experiences similar London-dispersion forces **(160amu)**
36. Which type of attraction bond has the least relative strength? **(London forces)**
37. Stronger intermolecular forces _____ viscosity. **(Higher)**
38. Surface tension is a measure of the toughness of the surface of a liquid. Stronger intermolecular forces _____ surface tension **(Higher)**

39. _____ is defined as the pressure exerted by a vapor in thermodynamics equilibrium with its condensed phase at a given temperature in a closed system **(both Vapor pressure or equilibrium vapor pressure)**
40. Stronger intermolecular forces _____ vapor pressure **(lower)**
41. _____ are molecules that allow organism to transfer genetic information from one generation to the next. **(Nucleic acid)**
42. _____ bears the hereditary information that's passed on from one parent to children **(DNA)**
43. In the nucleic acid if the sugar is a simple ribose, the polymer is **(RNA)**
44. In the nucleic acid of the sugar is derived from ribose as deoxyribose, the polymer is _____ which of the following **(RNA)**
45. The five carbon sugar is called _____ **(De-oxyribose)**
46. The second carbon of ribose bears a _____ group **(Hydroxyl)**
47. The second carbon of de-oxyribose has a _____ instead **(Hydrogen)**
48. Nucleic acids are found in abundance in all living things, where they functions _____ information in the nucleus of every living cell of every life-form organism on earth. **(to create, encode, store)**
49. Adenine and guanine and purines, meaning their structure contain _____ **(two fused carbon-nitrogen rings)**
50. Well-studied biological nucleic acid molecules range in size of 21 nucleotide (sample interfering RNA) to large chromosome human chromosome 1 is a single molecule that contains _____ million base pairs. **(247)**
51. Nuclein were discovered by Friedrich Miescher in _____ **(1869)**
52. In _____ Richard Alumann discovered the nuclein had acidic properties called nucleic acid **(1889)**
53. In 1938 _____ published the first X-ray diffraction pattern of DNA **(Astbury and Bell)**
54. In _____ Watsen and Crick determined the structure of DNA. **(1953)**
55. In eukaryotes such as plants and animals, DNA is found in the nucleus as well as in certain other types of organelles such as **(mitochondria, chloroplasts)**
56. In prokaryotes such as bacteria, the DNA is not enclosed in a membranes envelope although it's located in a specialized cell region called the **(Nucleoid)**
57. In eukaryotes, DNA is typically broken up into a number of very long, linear pieces called **(Chromosomes)**
58. In prokaryotes such as bacteria, chromosomes are much smaller and often **(Circular)**
59. A always pairs with T through two _____ bonds **(Hydrogen)**
60. The sugars and phosphates lie on the _____ of the helix, forming the backbone of the DNA; this portion of the molecule is sometimes called the sugar-phosphate backbone **(Outside)**
61. Organic compounds including fats, phospholipids, steroids and waves are **(Lipids)**

62. _____ percent of the human body is made up of protein **(20)**
63. Which type of protein contains prosthetic group **(Conjugated)**
64. The most common type of secondary structure is/are **(Alpha helix, Beta pleated sheets)**
65. The stress exerted on a body of rock by surrounding rock is a pressure in Earth's crust somewhat analogous to hydrostatic pressure in fluid is called **(Lithostatic pressure)**
66. Enthalpy is the energy transferred between a system and the surrounding under constant **(Pressure)**
67. The _____ structure is held together by covalent bonds such as peptide bonds **(Primary)**
68. Thermodynamics is the field of physics that deals with the relationship between heat and other properties such as _____ in a substance. **(Pressure, Density, Temperature)**
69. The heat of a material is understood as a representation of the energy contained within the particles of that material. This is known as the _____ theory of gases **(Kinetic)**
70. When a low-conducting material is used to prevent heat transfer. This is called _____ **(Insulation)**
71. _____ are these that are independent of the size of a system **(Intensive properties)**
72. _____ system is a system that freely exchanges energy with its surroundings **(Open)**
73. _____ system is a system that exchanges only energy with its surroundings, not matter **(closed)**
74. A process during which the pressure remains constant is called _____ **(Isobaric)**
75. The IUPAC recommendation for pressure is a _____ **(lower-case p)**
76. Atmospheric pressure is close to _____ pascal **(100,000)**
77. A device that measures pressure using a column of liquid is called a _____ **(manometer)**
78. The rolling rock has _____ **(Kinetic energy)**
79. The kinetic energy of a system of the kinetic, potential and internal energies: Expressed as **(U-KT+PT)**
80. The unit of measurement for enthalpy in SI is **(Joule)**
81. Enthalpy of a reaction is energy change of a reaction **Delta H**, is the amount of energy on heat absorbed in a reaction. If the energy is required, **delta H** is _____ **(Positive)**
82. The enthalpy change which occur when a mole of the compound is burned completely in oxygen under standard conditions, and with everything in its standard state, this is called **(The enthalpy of combustion)**
83. The enthalpy change of combustion will always have a _____ value **(Negative)**
84. Enthalpy of denaturation of Lysozyme _____ kcal/mole **(63.6) (option not confirmed)**

85. Enthalpy of _____ defined as the enthalpy change observed when one mole of gaseous ions is completely dissolved in water forming one mole of aqueous ions. **(Hydration)**
86. Enthalpy of _____ defined as the heat that is absorbed when melting occurs at constant pressure. **(Fusion)**
87. Enthalpy of _____ defined as the enthalpy change required to completely change the state of one mole of substance between solid and gaseous states **(sublimation)**
88. For sodium chloride, the solid is more stable than the gaseous ions by _____ kJ **(787)**
89. A _____ reaction is one that is able to proceed without needing an outside source of energy **(Spontaneous)**
90. Ice melting at -10 degree Celsius is an example of _____ **(Non- spontaneous)**
91. _____ is the process in which cells absorb molecules by engulfing them **(Endocytosis)**
92. The plasma membrane creates a small deformation inwards, called an _____ **(Invagination)**
93. When the dissolved material enters the cell it is called **(Phagocytosis)**
94. The thermal decomposition of copper carbonate is a _____ reaction **(spontaneous)**
95. The higher up an organism is on the food chain, the _____ available energy it receives/requires from its food sources. **(Less)**
96. In all spontaneous processes, the total entropy always increases and the process is _____ **(irreversible)**
97. In all first formulations created by the French scientist Sadi Carnot, who in _____ showed that there is an upper limit to the efficiency of heat to work, in a heat engine **(1824)**
98. The _____ is concerned with the direction of natural processes. It asserts that a natural process runs only in one sense and is not reversible **(second law)**
99. The _____ also predicts the end of the universe **(second law)**
100. _____ of thermodynamics was developed by the German chemist Walther Nernst during the years 1906 **(third law)**
101. The third law of thermodynamics is often referred to as **(Nernst's theorem, Nernst postulate)**
102. An alternative version of the third law of thermodynamics as stated by _____ in 1923 **(Gilbert Lewis, Merle Randall)**
103. The third law of thermodynamics states that the entropy of a system approaches a constant value as the temperature approaches _____ **(absolute zero)**
104. The entropy of a system at absolute zero is typically _____ and in all cases is determined only by the number of different ground states it had. **(Zero)**
105. It says that when we are considering a totally perfect (100%) crystalline structure, at absolute zero (0kelvin) it will have **(no entropy)**

106. Reactions that have a negative **delta G** release free energy and are called _____ reactions. **(Exergonic)**
107. Reaction with a positive delta G, on the other hand, require an input of energy and are called _____ relations **(Endergonic)**
108. Living system does not violate the _____ law of thermodynamics, which states the entropy increase over time. **(Second)**
109. Depending on the membrane's location and role in the body, lipids can make up anywhere from _____ percent of the membrane, with the remainder being proteins. **(20-80)**
110. The element _____ is a one-atom molecule **(Helium)**
111. The process of _____ a substance tends to move from an area of higher concentration to an area of lower concentration until its concentration become equal throughout a space. **(Diffusion)**
112. Kinetic Energy can be increase by increasing the _____ **(temperature)**
113. In _____ oxygen and carbon dioxide are not exchanged by diffusion in the lungs **(Mammals, Birds, Reptile and amphibians)**
114. _____ is the passive movement of molecules across the cell membrane via the aid of a membrane protein. **(Facilitated diffusion)**
115. How many types of membrane are possible **(three)**
116. At rest, the inside of the membrane is _____ charged, while the outside is _____ charged. **(Negatively, positively)**
117. If the water concentration of the cells cytoplasm is lower than that of the medium is a _____ solution **(hypotonic)**
118. Friction can be reduced by _____ to rubbing surface **(applying lubricants oil or grease)**
119. Gel filtration chromatography is a separation based on _____ **(Size)**
120. Gel filtration chromatography is also known as **(Size- exchange and molecular sieve chromatography)**
121. The Gel filtration chromatography technique was invented by _____ **(Grant Henry Lathe and Cline R Ruchven)**
122. Theodor Svedberg invented the analytical ultra-centrifuge in _____, and won the Nobel Prize in Chemistry in 1926 for his research on colloids and proteins using the ultracentrifuge. **(1923)**
123. The European X-ray free-election laser is an x-ray research laser facility commissioned during May _____ **(2017)**
124. Resolution of single-particle reconstruction of beta-galactosidase is _____ **(2.2 anstrong)**
125. Edward Mills Purcell and Felix Bloch shared the _____ Nobel Prize in Physics for their discoveries. **(1952)**

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